Anonymisation of the E-Voting Survey

Security and Privacy – Final Project, Phase 1 (Group C)

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Summary

We publish an anonymised version of a survey of 200 voters from a local election in which some people could vote online. In short:

- **Privacy.** We removed names, replaced exact birth dates with three broad age groups, and removed marital status, citizenship and ZIP code. Nothing had to be blanked out and no answer was altered. An attacker using only public information can expect to identify about 2 of the 200 people correctly, and can never be sure whether a given voter took part in the survey.
- **Usefulness.** We ran the same tests on the original data and on the released file. All 9 tests that can be repeated on the released file reach the same conclusion, including both findings that were significant in the original data. The 10 tests that need a removed variable were all non-significant in the original data.
- **Answers.** (A) The survey agrees with the official election results, for both paper and online votes. (B) Older and more highly educated voters were more likely to vote Red. (C) We found no clear difference in demographics between online and paper voters.

1 Introduction

In the latest local election, one municipality let its voters choose between voting online and voting on paper at a polling station. The online votes were distributed differently from the paper votes. To check for possible manipulation, and to learn who chooses to vote online, a survey asked 200 voters how they voted and recorded some facts about them. The survey was meant to answer three research questions:

(A) Do the survey answers differ from the official election results, for paper votes and for online votes?
(B) Does the choice of party depend on who the voter is (age, sex, education, . . .)?
(C) Does the choice to vote online depend on who the voter is?

We act as the statistical agency. We want to publish the survey so that others can check these analyses, without revealing how any identifiable person voted. Section 2 describes the data and who might misuse them. Section 3 explains what we did to protect the respondents, and Section 4 how much risk remains. Section 5 answers (A)–(C) using only the released file and the public election results, so anyone can reproduce the answers. Section 6 compares those answers with the same tests on the original data and discusses whether the trade-off is acceptable.

We used the original survey data for three things only: running the tests on the original data for comparison, choosing the protection settings, and the comparison in Section 6. Every number comes from one Python program that can be re-run.

2 The Data and Who Might Attack It

The survey. The original file has 200 records, one per respondent. Each record contains the person's name, sex, date of birth, ZIP code (one of four), education (9 levels), citizenship, marital status (4 levels), whether they voted online or on paper, and the party they voted for: Red, Green or an invalid ballot.

Public information. Two other files are public:

- a **population register** of the municipality's 1 527 residents. It lists name, sex, date of birth, ZIP code, citizenship, marital status, and how each person voted: on paper, online, or not at all. 1 069 of them voted;
- the **official results**: the number of Red, Green and invalid votes at each of the four polling stations, and among the online votes.

Three kinds of information. Privacy protection treats variables differently depending on what they reveal.

- A **direct identifier** identifies a person on its own. Here that is the *name*.
- **Quasi-identifiers** do not identify anyone on their own, but they can single a person out when combined and compared with public data. A woman born on a particular day and living in a particular ZIP code is very likely the only such person in the register. Here the quasi-identifiers are sex, date of birth, ZIP code, citizenship, marital status and voting channel, because all of them also appear in the public register.
- The **sensitive attribute** is what must be protected: the *party*.

Education is not in the register, so it cannot be used to look people up.

One detail matters a lot. For every respondent, the survey's "voted online or on paper" answer matches the register exactly. The voting channel is therefore effectively public. An attacker immediately knows that every respondent is one of the 1 069 voters, and which channel each of them used.

Who might attack the data? We consider two attackers.

- **Attacker 1** (the main assumption of the assignment) has the released file, the register, the official results and this report, but does *not* know who answered the survey. For each released record, Attacker 1 can count how many voters in the register look the same. If many voters match, the attacker cannot tell which of them the record belongs to. Because the register is public, we can count these matches exactly.
- **Attacker 2** (a stress test) also knows the list of the 200 respondents, for example because it leaked. Attacker 2 then knows each respondent's exact details from the register, and only the 200 respondents are candidates. That makes protection much harder. We test this case to see how the release would hold up if the list became known.

3 How We Protected the Respondents

We protected the data in five steps. Table 1 summarises what happened to each variable, and Table 2 in Section 4 shows how much each step reduced the risk.

Table 1: What happened to each variable.

Variable	Type	What we did
name	direct identifier	removed
date of birth	quasi-identifier	replaced by age on 1 July 2025, in 3 groups: 20–39, 40–59, 60+
ZIP code	quasi-identifier	removed
citizenship	quasi-identifier	removed
marital status	quasi-identifier	removed
sex, voting channel	quasi-identifier	kept unchanged
education	not public	9 levels merged into 4
party	sensitive	kept unchanged

Step 1: Remove the name and shuffle the rows. We deleted the name and put the records in random order, so that a record's position in the file gives nothing away.

Step 2: Make details less precise. An exact date of birth would make almost everyone unique, so we replaced it with the person's age on the survey date, grouped into three age bands. We merged education from nine levels into four standard levels, based on the international ISCED scale. Education is not public, so this is not a privacy measure. We did it because several of the original levels had fewer than five respondents, which is too few for the chi-square tests in Section 5.

Step 3: Remove marital status and citizenship. After step 2, most of the remaining risk came from these two variables. Small groups, such as widowed people or citizens of 16 different countries, make individuals easy to pick out. Removing them costs very little, because in the original data neither is significantly related to party or to voting channel: all corrected p-values (explained in Section 5) are above 0.2.

Step 4: Remove the ZIP code. We first merged the four ZIP codes into two areas. Even then, combining sex, age group, area and channel split the respondents into many small groups, and in some of them everyone had voted for the same party. An attacker who can place a person in such a group learns that person's vote without knowing which record is theirs. Removing the ZIP code halves the number of possible profiles, from 24 to 12, and leaves no group in which everyone voted the same way. ZIP code is not significantly related to party or channel in the original data. The cost is that question (A) can no longer be checked polling station by polling station; none of those checks was significant in the original data either.

Step 5: Check two rules. The release must satisfy two rules:

1. every record must look the same as at least 5 voters in the register. This is known as *k-anonymity* [3], here checked against the whole register;
2. every respondent must fit at least 2 released records, judged by their *exact* details as Attacker 2 would know them. Then no respondent can be pinned to a single record.

If a rule were broken, a simple computer procedure would blank out single values until both rules hold. After step 4, both rules already hold, so 0 values were blanked and the file has no empty cells. The only missing information is education for the 3 respondents who answered "Not stated". That label is kept in the file and left out of the education tests.

Why we did not alter any answers. An alternative protection is to change some party answers at random on purpose (post-randomisation, or PRAM). We tried it in an earlier version. With only 200 respondents, the amount of shuffling we could afford without changing the conclusions (about 10% of answers) left every single-party group intact, so it protected almost nothing. Removing the ZIP code solved the same problem completely, and every answer in the file stays true.

How we chose these settings. We tried 72 combinations: three ways of grouping age, four choices of which of marital status and citizenship to remove, ZIP codes merged or removed, and three strictness levels for the rules in step 5. For each combination we measured the risk (Section 4) and re-ran every test from Section 5 to see how many conclusions from the original data survive [1]. Figure 1 shows the result. The released file is the safest combination. It also keeps the fewest tests, but every test it loses was non-significant in the original data.

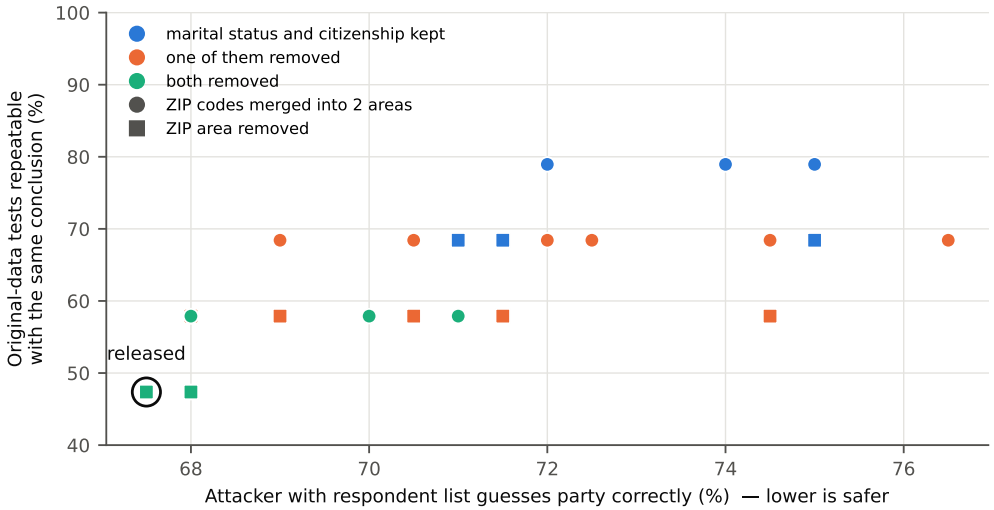


Figure 1: Each dot is one combination of protection settings that satisfies both rules from step 5. Further left means an attacker with the respondent list guesses party correctly less often (safer). Higher means more of the tests from the original data can be repeated with the same conclusion (more useful). The circled dot is the released file.

4 How Much Risk Remains

A record's *profile* is the combination of its released details that are also public: sex, age group and voting channel. We measure risk separately for the two attackers from Section 2.

Attacker 1 (no respondent list).

- *Records matching fewer than 5 voters*: how many records share their profile with fewer than five voters in the register. The goal is zero.
- *Expected correct re-identifications*: suppose that for every record the attacker picks one of the matching voters at random. This is the average number of the 200 picks that would be right.
- *Highest share of matching voters who took the survey*: for each profile, the fraction of matching voters who are respondents. If it is low, the attacker cannot even tell whether a given person took part.

Attacker 2 (has the respondent list).

- *Respondents unique within the survey*, and the *expected correct re-identifications* among the 200 respondents.
- *Respondents whose matching records all show one party.* For them the attacker learns the party without knowing which record is theirs (in technical terms, ℓ-diversity of 1 [2]).
- *Party guessed correctly*: we simulated the attack. For each respondent on the list, the attacker finds all released records the respondent could be and guesses their most common party. We score this twice: as the attacker would do it, and with the respondent's own record left out. The second score shows how much the attacker learns from the *other* respondents. For comparison, an attacker using only public data would guess the party that won the person's polling station (or the online vote), and would be right for 62.5% of respondents.

Table 2: Risk after each protection step (200 records). An attacker using only public data guesses the party correctly for 62.5% of respondents. Attacker 2's figures involve the party, so for the versions we did not release (steps 2 and 3) they are left out (–): they would hint at voting patterns by ZIP code or marital status.

After step:	1. Name removed	2. Details coarsened	3. Marital, citizenship removed	4. ZIP area removed (released)
Attacker 1: no respondent list				
Records matching fewer than 5 voters	200	26	0	0
Expected correct re-identifications (of 200)	199.5	21.9	4.6	2.2
Highest share of matching voters who took the survey	1.00	1.00	0.29	0.25
Attacker 2: has the respondent list				
Respondents unique within the survey	200	–	–	0
Expected correct re-identifications (of 200)	200.0	–	–	12.0
Respondents whose matching records all show one party	200	–	–	0
Party guessed correctly	100.0%	–	–	67.5%
same, with the respondent's own record left out	63.0%	–	–	66.0%

What the numbers show (Table 2).

- With only the name removed, *every* record is unique, mainly because of the exact date of birth.
- Making details less precise removes most of this uniqueness, but 26 records still match fewer than five voters.
- Removing marital status and citizenship protects against Attacker 1: every record now matches at least five voters. Attacker 2, however, could still pin some respondents to a single record, and some groups had all voted the same way.
- Removing the ZIP code solves both problems. Against Attacker 1, the expected number of correct re-identifications falls to 2.2 of 200 (1.1%), and at most a quarter of the matching voters for any profile took the survey. Against Attacker 2, no respondent is unique, no group is all one party, and every profile is shared by at least five respondents.
- The simulated attack still guesses the party correctly for 67.5% of respondents, against 62.5% with public data only. Leaving the respondent's own record out barely changes

this (66.0%), so the gain does not come from finding the person's record. It comes from a general pattern visible in the whole file: older voters vote Red more often (Section 5.2). Any release that can answer question (B) has to show this pattern.

To avoid giving away extra information, this report does not show any count below five. It also does not show exact results from the original data for groups an attacker can identify in the register (ZIP code, marital status, citizenship, exact age). With the respondent list, such numbers could be worked back into vote counts for those groups.

5 Answers to the Research Questions

How to read the results.

- We use two standard tests. The **chi-square goodness-of-fit test** checks whether observed counts match a known set of shares. The **chi-square test of independence** checks whether two variables are related, for example whether the share of Red voters differs between age groups. Both compare the counts we see with the counts we would expect if there were no difference. They are reliable when every expected count is at least 5, which holds for every test in this section.
- A **p-value** is the probability of seeing a difference at least as large as the observed one if there were really no difference. A value below 0.05 is called *statistically significant*.
- Running several tests means some would fall below 0.05 by chance alone. The **Bonferroni correction** multiplies each p-value by the number of tests for the same question, capped at 1. All p-values in the tables are corrected this way.
- **Cramér's** V measures how strong a relationship is, from 0 (none) to 1 (perfect).

Everything in this section was computed from the released file and the official election results only.

5.1 (A) Does the survey agree with the election results?

How we tested it. For each channel (paper and online), we compared the number of Red and Green votes in the survey with what the official shares predict, using the goodness-of-fit test. Invalid ballots are left out: only about two are expected per channel, too few for the test. They make up a similarly small share in the survey and in the official results (under 4% in both channels).

Table 3: (A) Survey compared with the official results, among valid votes.

	Valid votes in survey	Red in survey, %	Red, official %	χ^2 (1 df)	p-value (Bonferroni)	Significant?
Paper votes	144	36.1	34.5	0.16	1.000	no
Online votes	52	34.6	37.0	0.13	1.000	no

What we found. Neither the paper votes nor the online votes in the survey differ from the official results (Table 3). The Red shares are within about two percentage points of the official ones, and the corrected p-values are 1. The survey therefore gives no sign that the online count differs from what voters say they voted.

There is an important limit. A survey of this size can only reliably detect large differences: about 11 percentage points for paper votes and about 19 for online votes, where only 52 respondents gave a valid vote. Smaller manipulations would go unnoticed.

5.2 (B) Does party choice depend on demographics?

How we tested it. Among valid votes, we tested for each attribute whether the share of Red voters differs between its groups, using the test of independence.

Table 4: (B) How party (Red vs. Green) relates to each attribute.

Attribute	χ^2 (df)	p-value (Bonferroni)	Strength (Cramér's V)	Significant?
Sex	4.21 (1)	0.161	0.15	no
Age group	15.83 (2)	0.001	0.28	yes
Education	30.45 (3)	<0.001	0.40	yes
Voting channel	0.04 (1)	1.000	0.01	no

What we found (Table 4). Age and education are clearly related to party.

- The Red share rises with age: from about one in five voters aged 20–39 to about half of those aged 60 and over.
- The Red share is about one in five among voters with primary or upper-secondary education, and about three in five among those with higher education.
- Women voted Red somewhat more often than men ($p = 0.040$ before correction), but after the Bonferroni correction this is not significant ($p = 0.161$).
- Voting channel is not related to party.

5.3 (C) Does the choice to vote online depend on demographics?

How we tested it. We used the same test, with "voted online or on paper" in place of party.

Table 5: (C) How voting online relates to each attribute.

Attribute	χ^2 (df)	p-value (Bonferroni)	Strength (Cramér's V)	Significant?
Sex	1.51 (1)	0.659	0.09	no
Age group	4.12 (2)	0.382	0.14	no
Education	3.00 (3)	1.000	0.12	no

What we found (Table 5). No attribute is significantly related to the choice of voting online. Younger respondents voted online somewhat more often than older ones, but the difference is not significant ($p = 0.127$ even before correction). With 200 respondents, such a modest difference cannot be established.

6 Was the Trade-off Worth It?

How useful is the released file? We ran every test from Section 5 on the original data as well, using the most detailed coding the chi-square tests allow: 10-year age bands, the four ZIP codes, and marital status and citizenship. Table 6 puts each result next to the same test on the released file.

- All 9 tests that can be repeated on the released file reach the same conclusion.

- Both findings that were significant in the original data (age and education in question B) are significant in the released file too, and no new significant result appears.
- For sex, education and voting channel, the strength of each relationship (Cramér's V) is exactly the same, because these variables were not changed. The age relationship is somewhat weaker, because six age groups were merged into three, but it remains clearly significant.
- The 10 tests that need a removed variable cannot be repeated. None of them was significant in the original data, including the per-station checks for question (A). One of those station checks also falls slightly short of the rule of five expected votes, so it was less reliable anyway.

To avoid leaking information, Table 6 gives exact original-data results only where the coding is the same as in the released file. Elsewhere it shows only the conclusion (see Section 4).

Table 6: Every test on the original data next to the same test on the released file. Bonferroni-corrected p-values, with Cramér's V in brackets for (B) and (C).

	Test	Original data: p (Cramér's V)	Released file: p (Cramér's V)	Same conclusion?
(A)	Paper votes	1.000	1.000	yes
(A)	Online votes	1.000	1.000	yes
(A)	Paper, station 2100	not significant	not in release	–
(A)	Paper, station 2200	not significant	not in release	–
(A)	Paper, station 2300	not significant	not in release	–
(A)	Paper, station 2400	not significant	not in release	–
(B)	Sex	0.282 (0.15)	0.161 (0.15)	yes
(B)	Age group	significant	0.001 (0.28)	yes
(B)	Education	<0.001 (0.40)	<0.001 (0.40)	yes
(B)	Marital status	not significant	not in release	–
(B)	Citizenship	not significant	not in release	–
(B)	ZIP code	not significant	not in release	–
(B)	Voting channel	1.000 (0.01)	1.000 (0.01)	yes
(C)	Sex	1.000 (0.09)	0.659 (0.09)	yes
(C)	Age group	not significant	0.382 (0.14)	yes
(C)	Education	1.000 (0.12)	1.000 (0.12)	yes
(C)	Marital status	not significant	not in release	–
(C)	Citizenship	not significant	not in release	–
(C)	ZIP code	not significant	not in release	–

Our verdict. We think the trade-off is good. The release answers all three research questions with the same conclusions as the original data, and every answer in it is true: nothing was blanked or altered. Against Attacker 1, the attacker the assignment assumes, the risk is very low. Against Attacker 2, who knows who took part, no respondent can be pinned to a single record and no group reveals a party. The remaining gain over public data comes from the age pattern that question (B) is about. The price is the per-station check in question (A) and the tests on marital status, citizenship and ZIP code, none of which showed an effect.

What we learned.

1. With a small survey, removing a few variables protected better than altering answers at random. The amount of random change we could afford without losing conclusions was too small to help.

2. Risk must be checked with what the attacker actually knows. A blanked value hides information in the file, but an attacker with the respondent list still knows that person's exact details.
3. Classical disclosure control protects well against the attackers we described, but gives no guarantee against attackers with other knowledge. Differential privacy would give such a guarantee, but with 200 records the noise it needs would hide the effects in question (B).

References

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